Data Visualization with Seaborn

CODE

```
#Seaborn:
#Data Visualization Library Built on Top of Matplotlib to create attractive and beautiful visualizations
```

CODE

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

CODE

```
df = pd.read_csv("sales_dataset.csv")
```

CODE

```
print(df.head())
print(df.info())
print(df.describe())
```

CODE

```
print(df.to_string())
```

CODE

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
```

CODE

```
sns.lineplot(x="Order Date", y="Sales", data=df, marker='*')

plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")

plt.show()
```

CODE

```
sns.lineplot(x="Order Date", y="Sales", hue="Region", data=df)

plt.title("Sales by Region")
plt.show()
```

CODE

```
plt.figure(figsize=(12,6))
plt.xticks(rotation=45)
sns.set_style("darkgrid")
```

```
sns.lineplot(x="Order Date", y="Sales",
hue="Region",
style="Region",
markers=True,
dashes=True,
data=df)
```

CODE

```
sns.barplot(x="Region", y="Sales", data=df, ci=None)
plt.show()
```

CODE

```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.barplot(x="Region", y="Sales", data=df)
plt.title("Sales")

plt.subplot(1,2,2)
sns.barplot(x="Region", y="Profit", data=df)
plt.title("Profit")

plt.tight_layout()
plt.show()
```

CODE

```
df_melted = df.melt(id_vars="Region",
value_vars=["Sales", "Profit"],
var_name="Category",
value_name="Value")
```

CODE

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

sns.barplot(x="Region", y="Value", hue="Category", data=df_melted, ci=None)

plt.title("Region-wise Sales vs Profit")
plt.xlabel("Region")
plt.ylabel("Value")

plt.show()
```

CODE

```
sns.scatterplot(x="Sales", y="Profit", data=df, color='red')

plt.title("Sales vs Profit")
plt.xlabel("Sales")
```

```
plt.ylabel("Profit")

plt.show()
```

CODE

```
sns.scatterplot(x="Sales", y="Profit",
hue="Region",
data=df)
```

CODE

```
sns.scatterplot(x="Sales", y="Profit",
hue="Region",
style="Region",
data=df)
```

CODE

```
sns.countplot(x="Region", data=df)

plt.title("Count of Records by Region")
plt.xlabel("Region")
plt.ylabel("Count")

plt.show()
```

CODE

```
sns.countplot(y="Region", data=df)
```

CODE

```
corr = df.corr(numeric_only=True)
print(corr)
```

CODE

```
sns.heatmap(corr, annot=True)
plt.show()
```